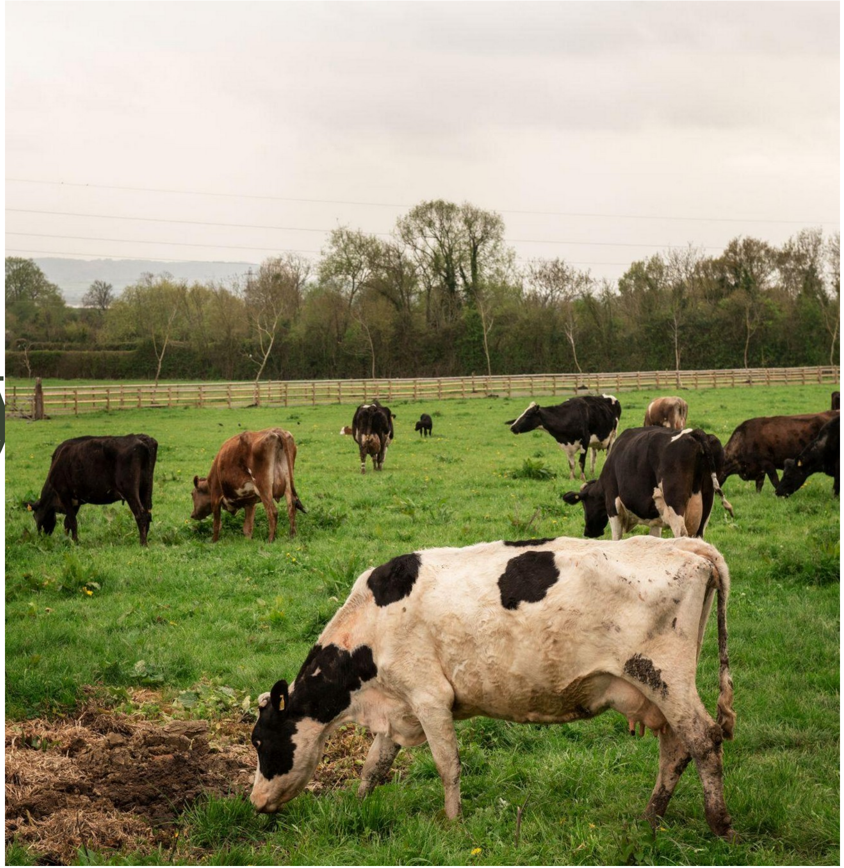
Strategic Analytics for Circular Dairy Bioeconomy

An Agent-Based Virtual Farm to for Sustainable Milk Production



The Problem: Why Dairy Farms Need Smarter Decisions

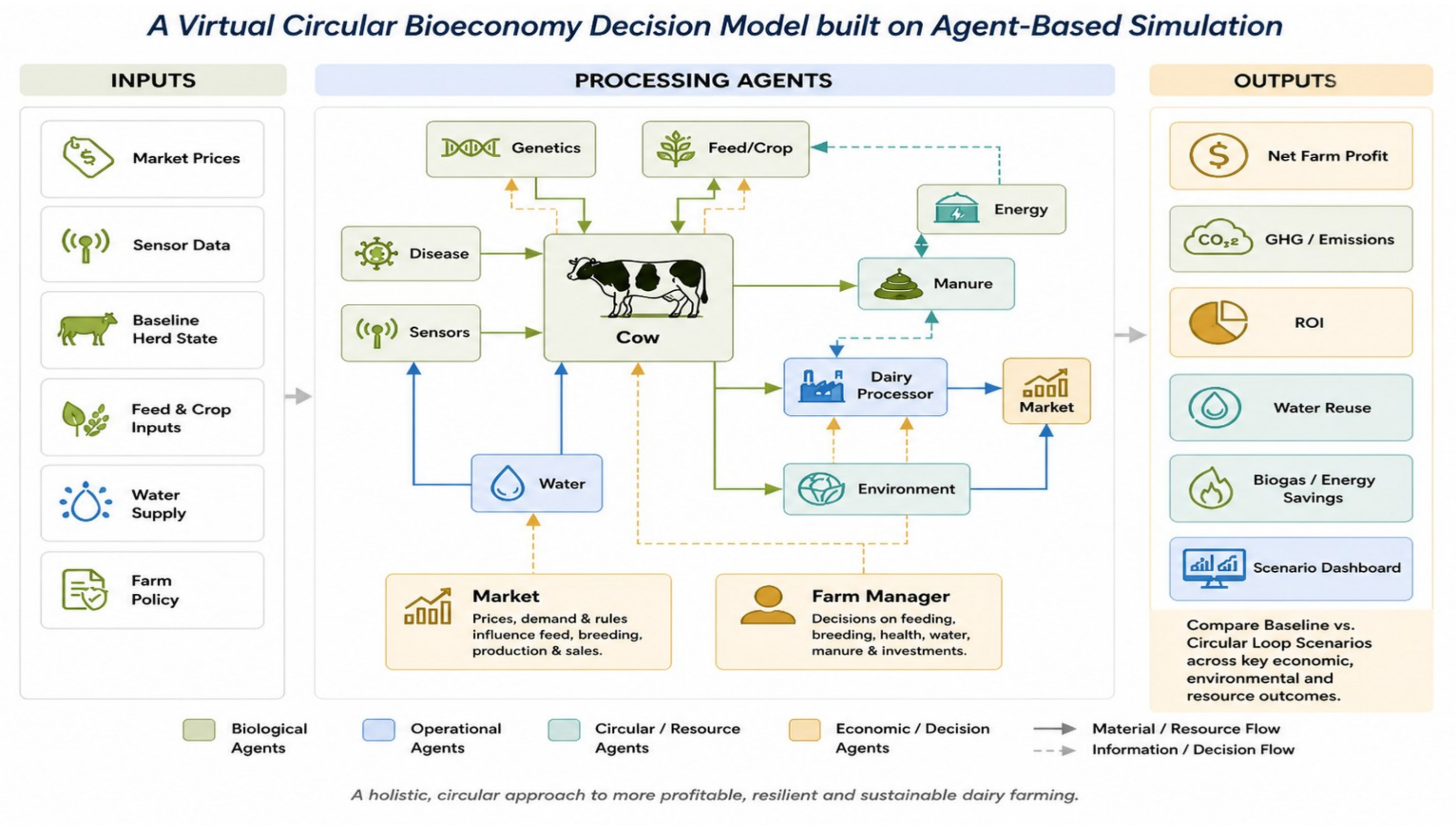
01. Dairy farms must improve **profitability and meet sustainability targets simultaneously**, with growing pressure from markets and regulators

02. Circular technologies like digesters, composting, and water recycling require **heavy capital,** but returns depend on **how they interact together** on each farm

03. Current tools like spreadsheets and single-purpose calculators **cannot capture** how nutrient, water, energy, and by-product strategies **interact across a whole farm**

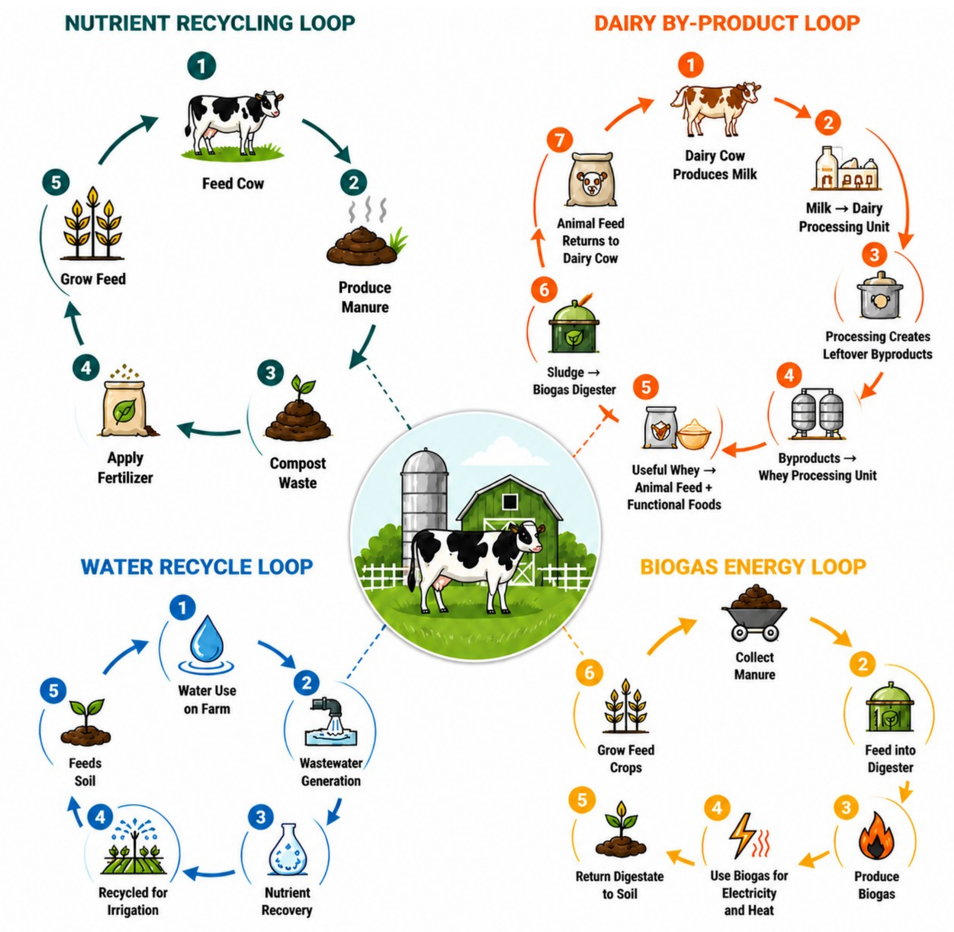
04. Without better tools, cooperatives cannot identify **highest-ROI investments**, estimate **payback periods**, or report **sustainability outcomes** to ESG stakeholders

The Solution: ViCBio DM Model

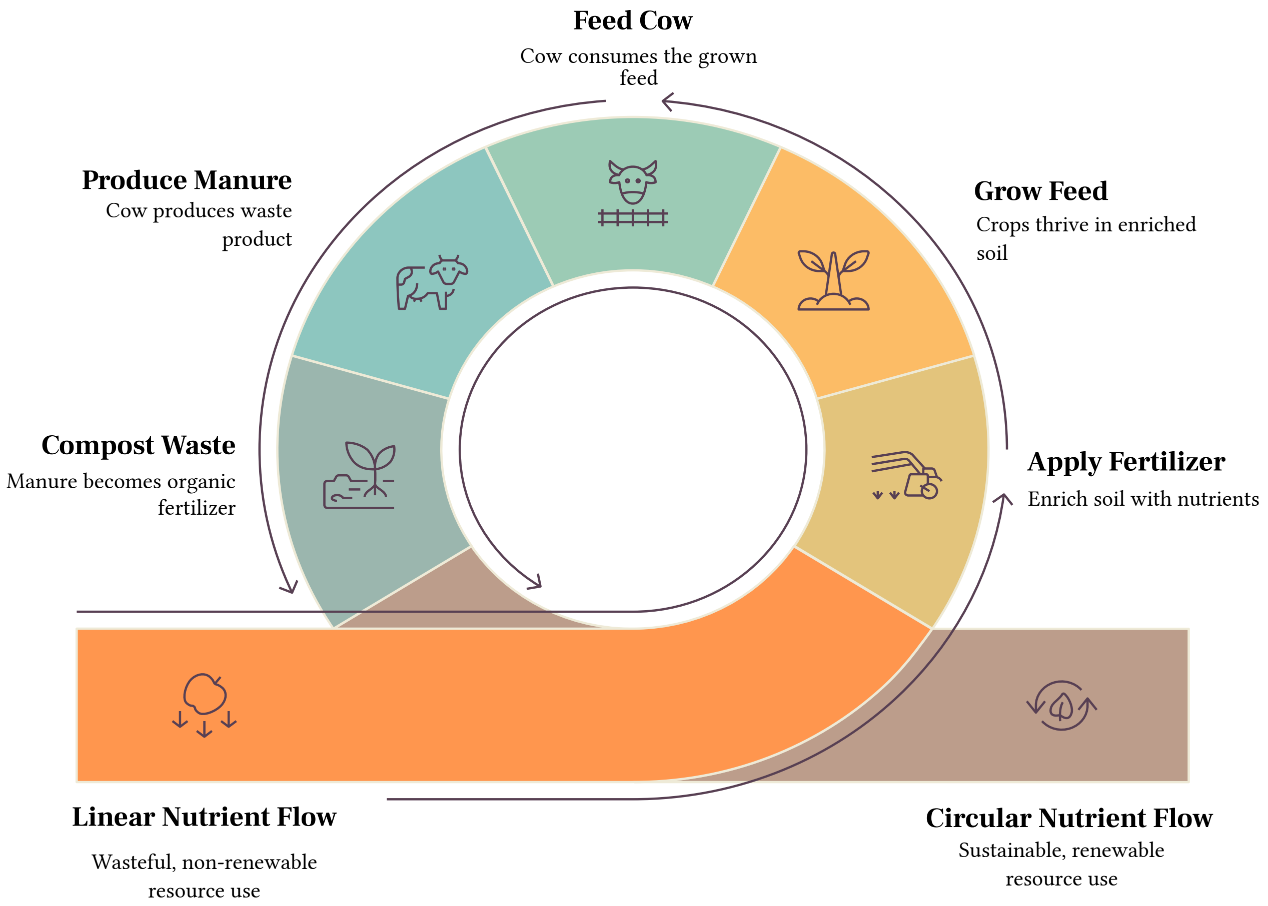


Circular Bioeconomy Loops

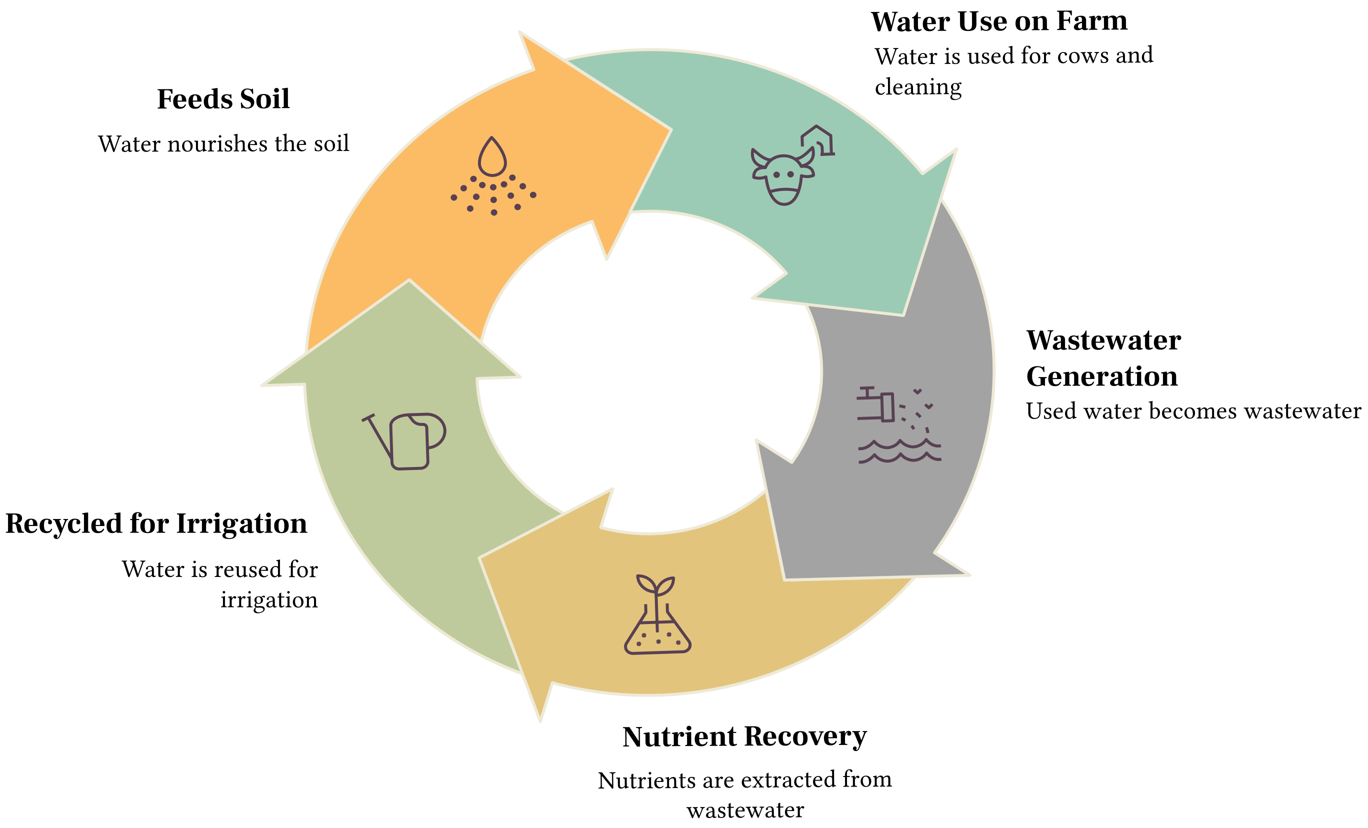
Loop	Business Value
Nutrient	Manure returns as fertilizer **cuts input costs**, builds **soil health**
Water	Wastewater treated and reused on-farm - reduces **freshwater demand**
Energy	Manure converts to **biogas** powers the farm and generates **new revenue**
By-product	**Whey and waste milk** redirected to **feed or biogas** nothing disposed



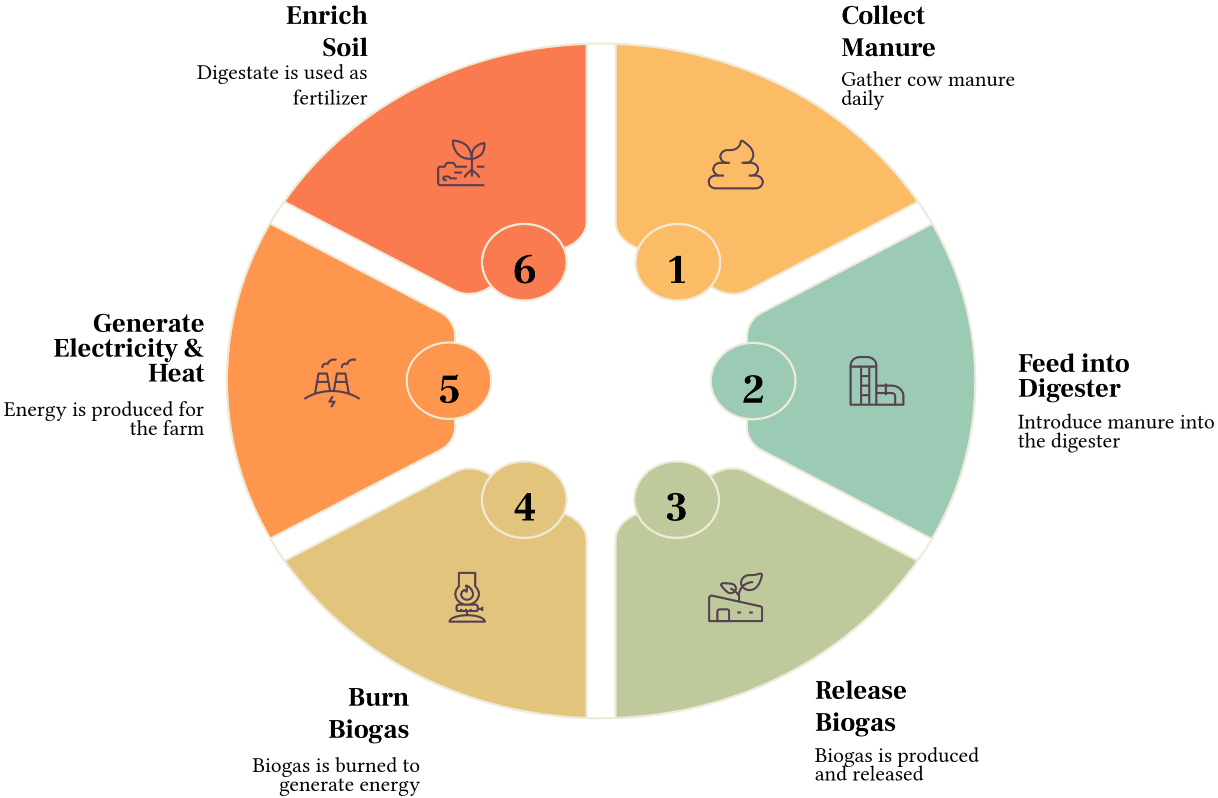
Nutrient Recycling Loop



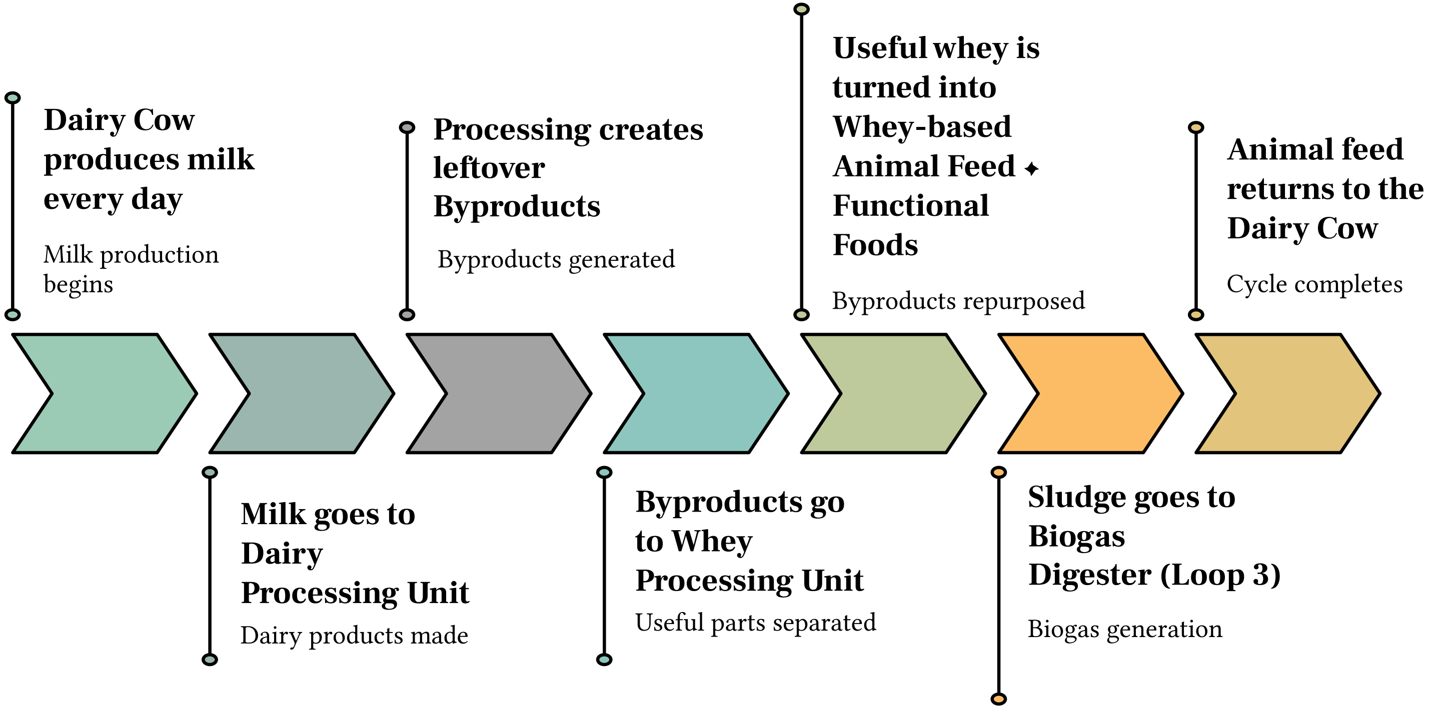
Water Recycle Loop



Biogas Energy Loop



Dairy By-product Cycle



Track Every KPI That Matters - In One View

Revenue Summary — Simulation Period

100 cows · 5.0 yr (1,825 days)

GROSS REVENUE
$3.991M
before input costs

NET PROFIT
$3.403M
revenue − feed & water

MILK REVENUE
$3.864M
96.8% of gross
$2.1K/day avg

ENERGY REVENUE
$54.9K
1.4% of gross
$30.08/day avg

HEAT VALUE
$18.9K
0.5% of gross
$10.35/day avg

COMPOST REVENUE
$53.2K
1.3% of gross
$29.17/day avg

Milk	96.8%
Energy	1.4%
Heat	0.5%
Compost	1.3%

INPUT COSTS DEDUCTED

Feed Cost
−$577.6K
−$316.51/day avg

Water Cost
−$10.2K
−$5.60/day avg

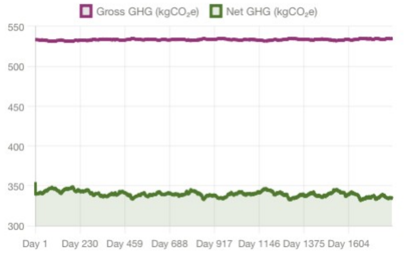
Farm Overview

Run the simulation from the top bar to see results. Adjust parameters in the sidebar first.

Avg. Daily Milk	**2714.1** L/day
Dairy Products Revenue	**$2.1K** $/day
Daily Whey Generated	**955.4** L/day
Energy Revenue	**$30.08** $/day
Heat Value	**$10.35** $/day
Compost Revenue	**$29.17** $/day
Net GHG/day	**339.6** $kgCO_2e$
Daily Electricity	**138.0** kWh/day
Daily Biogas	**69.0** m^3/day
Net Feed/day	**1438.7** kg/day
Daily Compost	**583.3** kg/day
Feed Conv. Ratio	**0.516** kg/kg

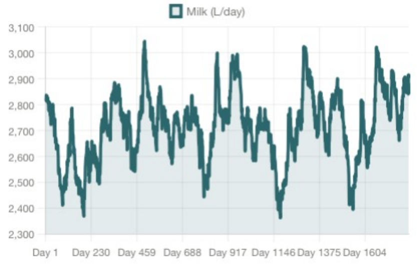
GHG Emissions

Gross vs Net kg CO_2e per day



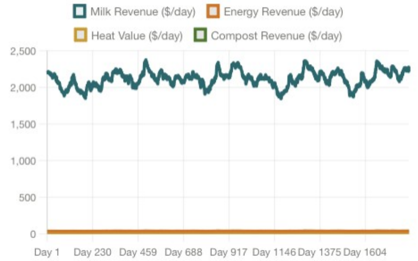
Daily Milk Production

Litres per day over simulation period



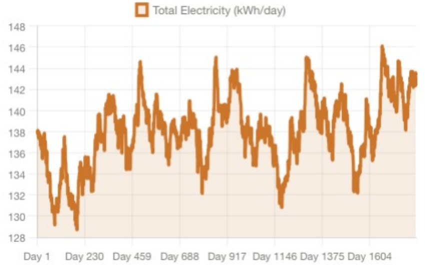
Daily Revenue

Milk, energy, heat & compost — separate streams



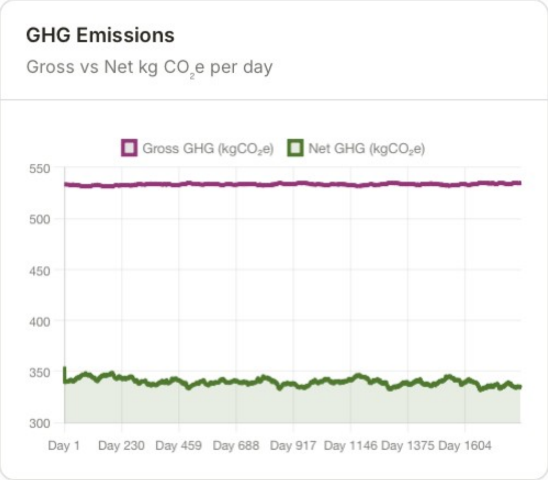
Energy Generation

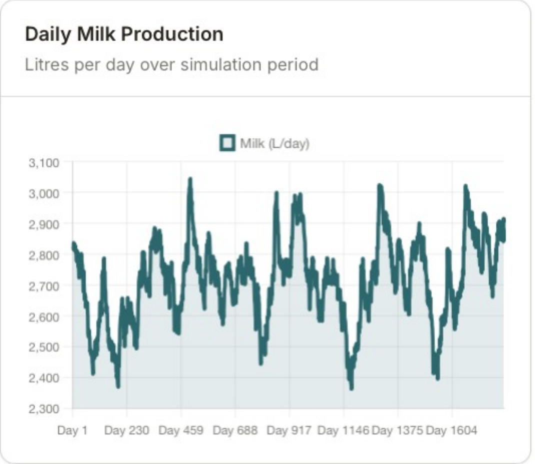
Biogas + solar electricity kWh/day

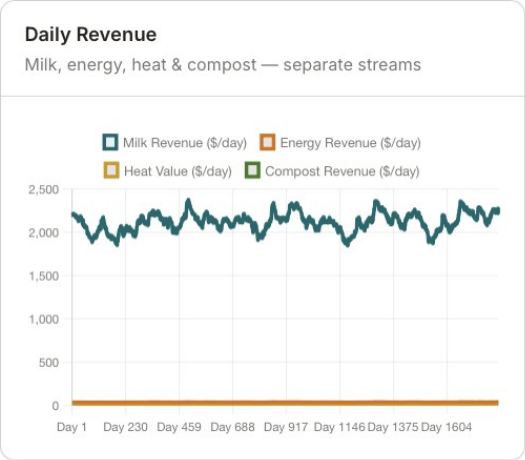


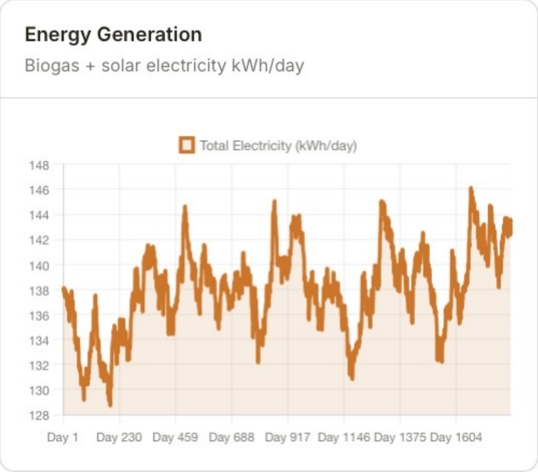
Simulated Daily Performance

Every critical farm metric, from milk production and feed efficiency to energy generation, water recycling, nutrient recovery, and GHG emissions, tracked daily across the entire simulation period.









Each Circular Loop Has a Measurable Return

LOOP 1

Manure Separator

Solid fraction → compost + N/P/K → cropland → feed offset

3048.0 kg/day

Total organic fertilizer to cropland

Feed offset from nutrients	**198.2 kg/day**
Nitrogen recovered	**7.29 kg/day**
Phosphorus recovered	**2.92 kg/day**
Potassium recovered	**8.75 kg/day**
Digestate applied	**2437.5 kg/day**

LOOP 2

Water Cycle

Wastewater treatment → irrigation → fresh-water offset

6,446 L/day

Recycled irrigation water

Wastewater treated	**9,209 L/day**
Fresh water offset	**4,512 L/day**
Water use per cow	**11510.7 L/day**

LOOP 3

Energy / Biogas

Manure → anaerobic digester → electricity + heat

138.0 kWh/day

Electricity generated

Biogas produced	**69.0 m³/day**
Heat recovered	**172.5 kWh/day**
Digestate → fertilizer	**2437.5 kg/day**

LOOP 4

Dairy Byproduct

Whey + waste milk → animal feed + functional foods

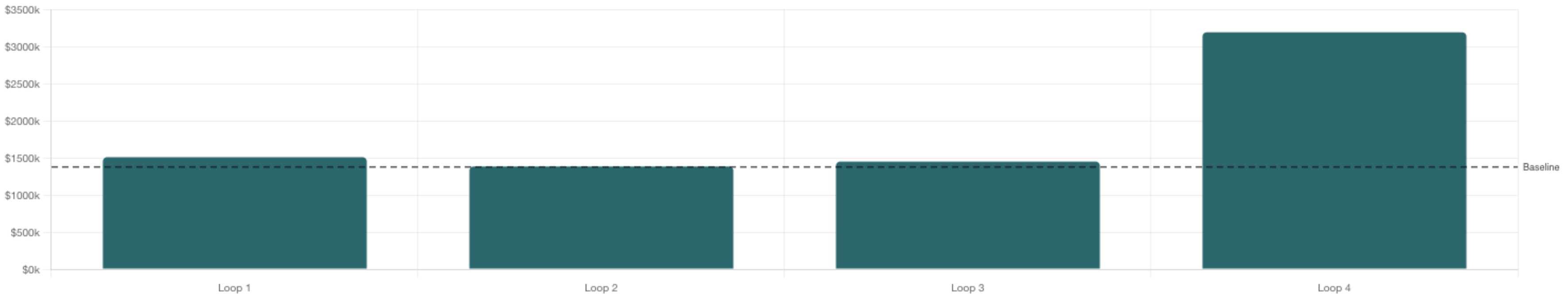
446.7 kg/day

Feed return from byproducts

Whey → functional foods	**501.4 kg/day**
Sludge → fertilizer	**27.1 kg/day**

Net Profit vs Baseline

Profit | Revenue | Feed Savings | Water Savings | GHG

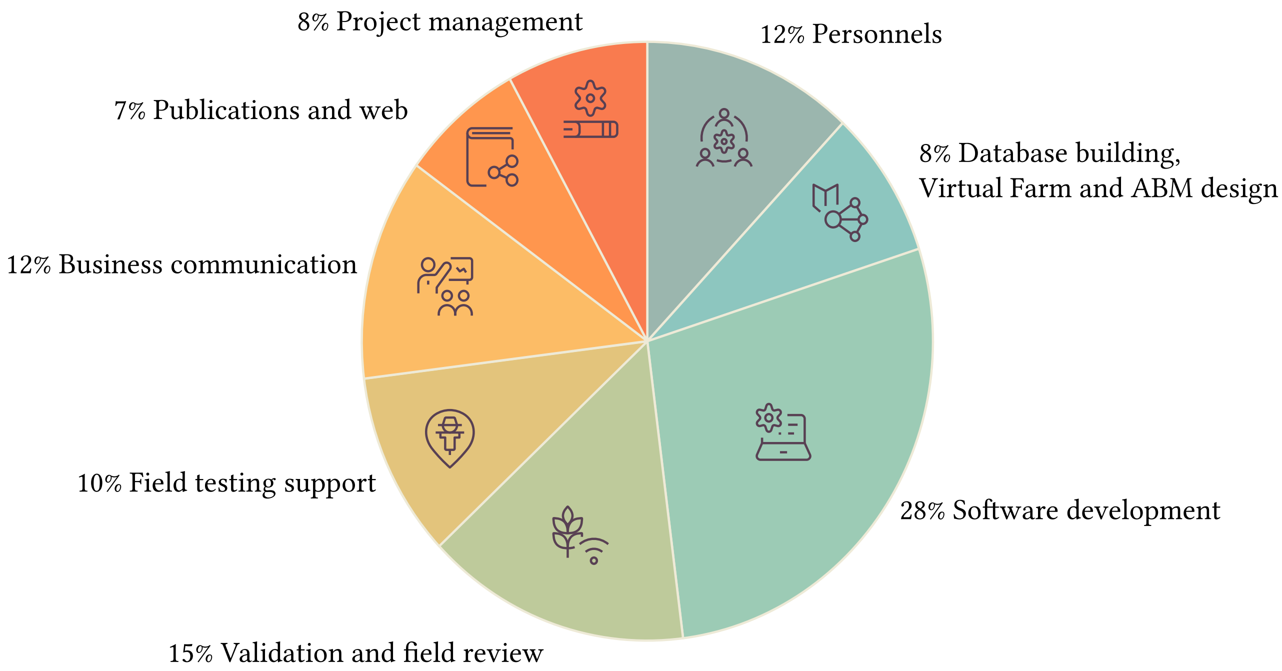


Equipment Capital Investment and ROI Analysis

Equipment (Herd size of 2000 cows)	Circular Loop	Total CapEx	Annual Net Benefit	Payback	Annual ROI	Lifetime ROI (15yr)	NPV @ 15yr
Anaerobic Biodigester	Energy Loop	$1,610,000	$247,200/yr	6.5 yrs	+15.3%	+130%	+$641,100
Biogas CHP Engine	Energy Loop	$689,500	$291,600/yr	2.4 yrs	+42.3%	+534%	+$1,970,000
Manure Separator	Nutrient Loop	$414,000	$855,800/yr	0.5 yrs	+204.4%	+3,000%	+$7,380,000
Water Recycling System	Water Loop	$506,000	$146,600/yr	3.5 yrs	+29.0%	+335%	+$829,200
Solar PV Array	Energy Loop	$805,000	$49,600/yr	16.2 yrs	+6.2%	-8%	-$353,200
Whey Processing Unit	Dairy Loop	$644,000	$1,170,000/yr	0.6 yrs	+181.2%	+2,618%	+$9,990,000
Smart Feed & Health Sensors	Precision Tech	$276,000	$914,500/yr	0.3 yrs	+331.3%	+4,870%	+$8,050,000
Portfolio Total	**All 4 Loops**	**$4,940,000**	**$3,670,000/yr**	**1.3 yrs**			**+$28,500,000**

Strategic Budget Allocation

Total funding distributed across six activity areas



What This Investment Delivers

Capital Planning

Compare digester, compost, water, and processing options with quantified ROI - before producer rollout

Risk Reduction

Identify low-performing strategies early and focus investment on combinations with the strongest returns

Sustainability Reporting

Generate defensible GHG and resource-efficiency metrics linked directly to operational choices ready for ESG stakeholders and customers

Market Differentiation

Position the cooperative as a leader in data-driven circular dairy innovation ahead of the industry curve

The dairy industry is ready. The technology exists. What's been missing is the decision-support to connect them until now.